

Memo Sem test 2 2011 WTW15f

SECTION A: MULTIPLE CHOICE QUESTIONS - 15 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 1 HOUR.
2. Use a soft pencil.
3. Fill in your personal information in pencil on the optic reader form on **SIDE 2**
4. Fill in your student number from top to bottom and then code it.
5. Answer questions 1 to 15 on the optic reader form on **SIDE 2**.
If side 1 is used it cannot be marked.
6. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
7. Do all scribbling on the facing page. It will not be marked.
8. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 15 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 1 UUR INHANDIG.
2. Gebruik 'n sagte potlood.
3. Vul u persoonlike inligting in potlood in op die merkleesvorm se **KANT 2**.
4. Vul u studentenommer van bo na onder in en kodeer dit.
5. Beantwoord vrae 1 tot 15 op die merkleesvorm se **KANT 2**
Indien kant 1 gebruik word kan dit nie nagesien word nie.
6. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
7. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
8. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

$$\lim_{x \rightarrow \infty} \frac{4x^3 + 5x^2 + 1}{2x^3 - 7x^2 + 4x}$$

[1a] = 2	[1b] = 0	[1c] = ∞	[1d] = $-\infty$
[1e] None of these / Geen van hierdie			

$$= \lim_{x \rightarrow \infty} \frac{4 + \frac{5}{x} + \frac{1}{x^3}}{2 - 7/x + 4/x^2}$$

$$= \frac{4}{2} = 2$$

Question 2 / Vraag 2

$$\lim_{x \rightarrow -\infty} \frac{4x^3 + 5x + 1}{2x^2 - 7x + 4}$$

[2a] = 2	[2b] = 0	[2c] = ∞	[2d] = $-\infty$
[2e] None of these / Geen van hierdie			

$$= \lim_{x \rightarrow -\infty} \frac{4x + 5/x + 1/x^3}{2 - 7/x + 4/x^2}$$

$$= -\infty$$

Question 3 / Vraag 3

$$\lim_{x \rightarrow -2} \frac{x}{|x+2|}$$

[3a] = 1	[3b] = 0	[3c] = ∞	<u>[3d] = $-\infty$</u>
[3e] None of these / Geen van hierdie			

as $x \rightarrow -2$ $\frac{x}{|x+2|}$ is negative as $x \rightarrow -2$ and $|x+2|$ is positive

Question 4 / Vraag 4

$$\lim_{x \rightarrow -\infty} e^{2x}$$

[4a] = 1	<u>[4b] = 0</u>	[4c] = ∞	[4d] = $-\infty$
[4e] None of these / Geen van hierdie			



Use the information below to answer Questions 5 and 6:

The distance (in meters) travelled by a car after t seconds is given by the function $s = f(t) = 2t^2$, $t \in [0, 10]$.

Gebruik die inligting hieronder om Vrae 5 en 6 te beantwoord:

Die afstand (in meter) afgelê deur 'n motor na t sekondes word gegee deur die funksie $s = f(t) = 2t^2$, $t \in [0, 10]$.

Question 5 / Vraag 5

The average velocity of the car over the interval $[1, 4]$ is

Die gemiddelde snelheid van die motor oor die interval $[1, 4]$ is

$$\frac{f(4) - f(1)}{4 - 1} = \frac{2 \times 16 - 2}{3} = \frac{30}{3}$$

[5a] $\frac{2}{3}$ m/s	<u>[5b] $\frac{30}{3}$ m/s</u>	[5c] $\frac{32}{3}$ m/s	[5d] None of these / Geen van hierdie
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Question 6 / Vraag 6

The (instantaneous) velocity of the car at time $t = 2$ is

Die (oombliklike) snelheid van die motor op tydstip $t = 2$ is

$$f'(t) = 4t \\ f'(2) = 8$$

<u>[6a] 8 m/s</u>	[6b] 10 m/s	[6c] 12 m/s	[6d] None of these / Geen van hierdie
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Question 7 / Vraag 7

The largest interval on which the function $f(x) = x^2 + 1 + \sqrt{x}$ is continuous, is

Die grootste interval waarop die funksie $f(x) = x^2 + 1 + \sqrt{x}$ kontinu is, is

[7a] $(-\infty, \infty)$	<u>[7b] $[0, \infty)$</u>	[7c] $(0, \infty)$	[7d] None of these / Geen van hierdie
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on its domain, that is for all x , $x \geq 0$

Question 8 / Vraag 8 $f'(x) = 2x + \frac{1}{2\sqrt{x}}$ and its domain is all $x > 0$

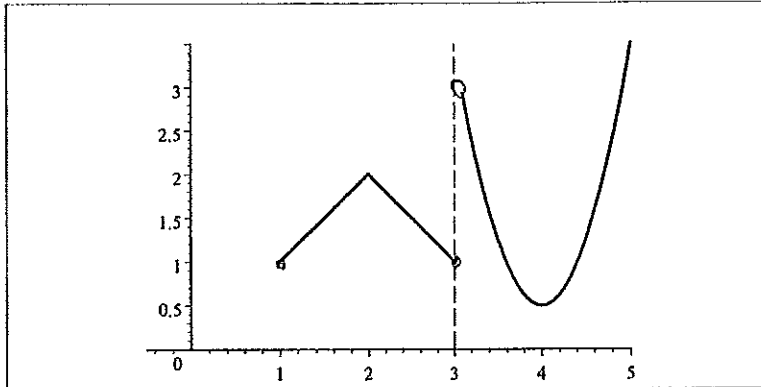
The largest interval on which the function $f(x) = x^2 + 1 + \sqrt{x}$ is differentiable, is

Die grootste interval waarop die funksie $f(x) = x^2 + 1 + \sqrt{x}$ differensieerbaar is, is

- | | | | |
|--------------------------|--------------------|--------------------------------------|---------------------------------------|
| [8a] $(-\infty, \infty)$ | [8b] $[0, \infty)$ | [8c] $(0, \infty)$ | [8d] None of these / Geen van hierdie |
|--------------------------|--------------------|--------------------------------------|---------------------------------------|

Use the graph of the function f below to answer Questions 9 and 10:

Gebruik die grafiek van die funksie f hieronder om Vrae 9 en 10 te beantwoord:



Question 9 / Vraag 9

Let $a \in \{2, 3\}$. The function f is continuous at $x = a$

Laat $a \in \{2, 3\}$. Die funksie f is kontinuu in $x = a$

- | | |
|--|---|
| [9a] only for $a = 2$ / slegs vir $a = 2$ | [9b] for $a = 2$ and $a = 3$ / vir $a = 2$ en $a = 3$ |
| [9c] only for $a = 3$ / slegs vir $a = 3$ | [9d] None of these / Geen van hierdie |

Question 10 / Vraag 10

Let $a \in \{2, 4\}$. The function f is differentiable at $x = a$

Laat $a \in \{2, 4\}$. Die funksie f is differensieerbaar in $x = a$

- | | |
|---|--|
| [10a] only for $a = 2$ / slegs vir $a = 2$ | [10b] for $a = 2$ and $a = 4$ / vir $a = 2$ en $a = 4$ |
| [10c] only for $a = 4$ / slegs vir $a = 4$ | [10d] None of these / Geen van hierdie |

Question 11 / Vraag 11

The equation of the tangent line to the graph of $f(x) = \sqrt{1-x}$ at $x = -3$ is

Die vergelyking van die raaklyn aan die grafiek van $f(x) = \sqrt{1-x}$ by $x = -3$ is

- | | | |
|---------------------------------------|--|--|
| [9a] $y + 3 = \frac{1}{4}(x - 2)$ | [9b] $y + 3 = -\frac{1}{4}(x - 2)$ | [9c] $y - 2 = \frac{1}{2\sqrt{1-x}}(x + 3)$ |
| [9d] $y - 2 = \frac{1}{4}(x + 3)$ | [9e] $y - 2 = -\frac{1}{4}(x + 3)$ | [9f] $y - 2 = \frac{-1}{2\sqrt{1-x}}(x + 3)$ |
| [9g] None of these / Geen van hierdie | | |

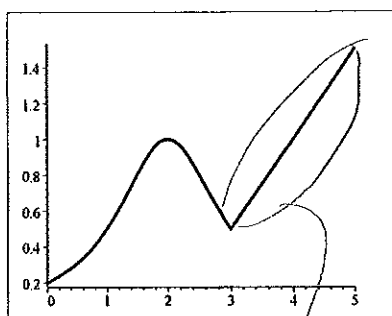
$$f'(x) = \frac{1}{2}(1-x)^{-1/2} (-1) = \frac{-1}{2\sqrt{1-x}} \quad f'(-3) = \frac{-1}{2\sqrt{4}} = -\frac{1}{4}$$

$$f(3) = \sqrt{4} = 2 \quad y - f(-3) = f'(-3)(x - (-3)) \quad y - 2 = -\frac{1}{4}(x + 3)$$

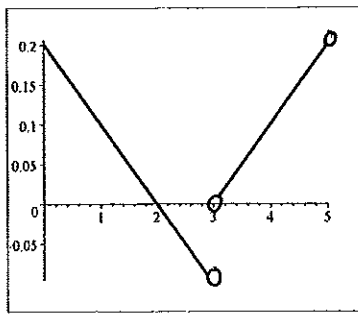
Question 12 / Vraag 12

Consider the graphs below. The derivative of the function $y = f(x)$ is the function

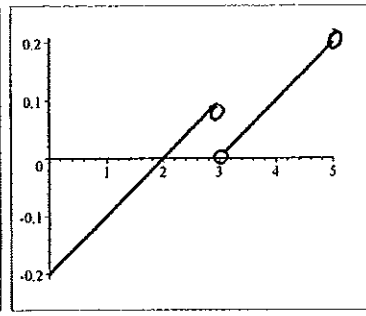
Beskou die grafieke hieronder. Die afgeleide van die funksie $y = f(x)$ is die funksie



$y = f(x)$

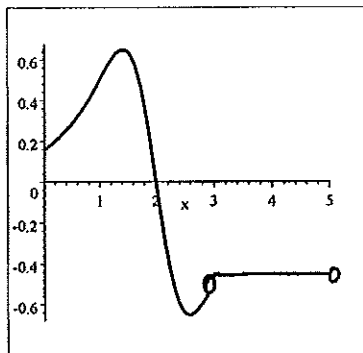


$y = g_1(x)$

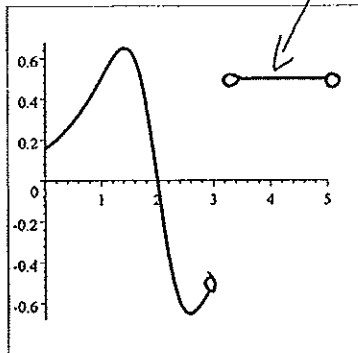


$y = g_2(x)$

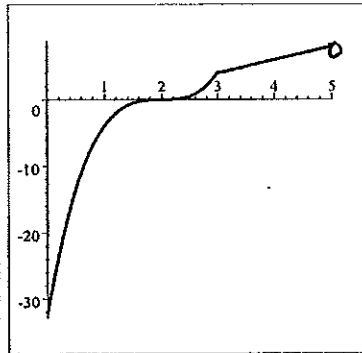
constant function, slope in a positive number



$y = g_3(x)$



$y = g_4(x)$



$y = g_5(x)$

[12a] $y = g_1(x)$	[12b] $y = g_2(x)$	[12c] $y = g_3(x)$	[12d] $y = g_4(x)$	[12e] $y = g_5(x)$
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Question 13 / Vraag 13

The equation(s) of the vertical asymptote(s) of the function $f(x) = \frac{x}{x^2 + 4x}$ is (are)

Die vergelyking(s) van die vertikale asimptoot(tote) van die funksie $f(x) = \frac{x}{x^2 + 4x}$ is

[13a] $x = 0$	[13b] $x = -4$	[13c] $x = 0$ and / en $x = -4$
[13d] $y = 0$	[13e] $y = -4$	[13f] $y = 0$ and / en $y = -4$
[13g] None of these / Geen van hierdie		

$$f(x) = \frac{x}{x(x+4)} = \frac{1}{x+4}, x \neq 0$$

$$\lim_{x \rightarrow 0} f(x) = \frac{1}{4} \quad x=0 \text{ is not a VA}$$

$$\lim_{x \rightarrow -4} f(x) = -\infty \quad x = -4 \text{ is a VA}$$

Question 14 / Vraag 14 $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{1}{x-1} = 0$ / $\lim_{x \rightarrow -\infty} f(x) = 0$

The equation(s) of the horizontal asymptote(s) of the function $f(x) = \frac{x}{x^2-1}$ is (are)

X Die vergelyking(s) van die horisontale asimptoot(tote) van die funksie $f(x) = \frac{x}{x^2-1}$ is

[14a] $x = 1$	[14b] $x = -1$	[14c] $x = 0$	[14d] $x = 1$ and / en $x = -1$
[14e] $y = 1$	[14f] $y = -1$	[14g] $y = 0$	[14h] $y = 1$ and / en $y = -1$
[14i] None of these / Geen van hierdie			

Question 15 / Vraag 15

Consider the function $f(x) = x + |x^3|$.

Consider the following statements:

Statement I: f is continuous at $x = 0$

Statement II: f is differentiable at $x = 0$

Which statement(s) is (are) true?

Beskou die funksie $f(x) = x + |x^3|$.

Beskou die volgende bewerings:

Bewering I: f is kontinu vir $x = 0$

Bewering II: f is differensieerbaar vir $x = 0$

Watter bewering(s) is waar?

$$f(x) = \begin{cases} x + x^3 & \text{if } x \geq 0 \\ x - x^3 & \text{if } x < 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 0$$

f is continuous at $x = 0$

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h + h^3 - 0}{h}$$

$$= \lim_{h \rightarrow 0^+} (1 + h^2) = 1$$

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{h - h^3 - 0}{h}$$

$$= \lim_{h \rightarrow 0^-} (1 - h^2) = 1 \therefore f'(0) = 1$$

[15a] only / slegs I	[15b] I and II / I en II
[15c] only / slegs II	[15d] None of these / Geen van hierdie

[15]

SECTION B: 30 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
If you need more space for an answer, use the facing page and indicate it clearly by framing it.
5. Show all steps and calculations and explain your answers.

AFDELING B: 30 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is word nie nagesien nie.
3. U mag nie korreger-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
5. Toon alle stappe en berekeninge en verduidelik u antwoord.

Question 16 / Vraag 16

Calculate the following limits.

Clearly show how you use theorems to find the limit.

You are not allowed to use l'Hospital's rule.

If you write down only an answer you will get no marks.

Bepaal die volgende limiete.

Toon **duidelik** aan hoe u stellings gebruik om die limiet te bepaal.

U mag nie die reël van l'Hospital gebruik nie.

Indien u slegs 'n antwoord neerskryf sal u geen punte kry nie.

$$\begin{aligned} \text{i} \quad \lim_{x \rightarrow 4} \frac{1 - e^{4-x}}{x+4} &= \frac{\lim_{x \rightarrow 4} (1 - e^{4-x})}{\lim_{x \rightarrow 4} (x+4)} \\ &= \frac{1 - 4^0}{4+4} = \frac{0}{8} = 0 \end{aligned}$$

[1]

$$\begin{aligned} \text{ii} \quad \lim_{x \rightarrow \infty} \frac{2e^x}{1+e^x} \quad & \left(\text{form } \frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow \infty} \frac{2}{\frac{1}{e^x} + 1} = \lim_{x \rightarrow \infty} \frac{2}{1} = 2 \end{aligned}$$

[1]

$$\text{iii} \quad \lim_{x \rightarrow -\infty} \tan\left(\frac{1}{x}\right).$$

$$= \tan\left(\lim_{x \rightarrow -\infty} \frac{1}{x}\right) = \tan 0$$

$$= 0$$

[1]

$$\text{iv} \quad \lim_{x \rightarrow 1} \frac{x-1}{x-\sqrt{x}} \quad \left(\text{form } \frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{x-1}{x-\sqrt{x}} \times \frac{x+\sqrt{x}}{x+\sqrt{x}} =$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+\sqrt{x})}{x^2 - x} = \lim_{x \rightarrow 1} \frac{(x-1)(x+\sqrt{x})}{x(x-1)}$$

[2]

$$= \lim_{x \rightarrow 1} \frac{x+\sqrt{x}}{x} = \frac{1+1}{1} = 2$$

$$v \quad \lim_{x \rightarrow 0} \frac{10x^5 + 6x}{x}$$

$$= \lim_{x \rightarrow 0} (10x^4 + 6)$$

$$= \lim_{x \rightarrow 0} 10x^4 + \lim_{x \rightarrow 0} 6$$

$$= 10 \times 0 + 6 = 6$$

[2]

X

$$vi \quad \lim_{x \rightarrow 2} \frac{4x \sin(x-2)}{(x-2)^2} \quad (\text{form } \frac{0}{0})$$

$$= \lim_{x \rightarrow 2} 4 \times x \times \frac{\sin(x-2)}{x-2}$$

$$= 4 \times \lim_{x \rightarrow 2} x \times \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x-2}$$

$$= 4 \times 2 \times 1 = 8$$

[3]

Question 17 / Vraag 17

Find the following derivatives. DO NOT SIMPLIFY YOUR ANSWER.

Bepaal die volgende afgeleides. MOENIE U ANTWOORD VEREENVOUDIG NIE.

i Find $\frac{d}{dx} \tan(x^3)$ / Bepaal $\frac{d}{dx} \tan(x^3)$

$$\frac{d}{dx} \tan x^3 = \sec^2(x^3) \times 3x^2$$

[1]

ii Find $\frac{dy}{dt}$ if $y = e^{3t}$ / Bepaal $\frac{dy}{dt}$ as $y = e^{3t}$

$$\frac{dy}{dt} = \frac{d}{dt} e^{3t} = 3e^{3t}$$

[1]

iii Find $\frac{d}{dx} \sqrt{x} \sec\left(\frac{1}{x}\right)$ / Bepaal $\frac{d}{dx} \sqrt{x} \sec\left(\frac{1}{x}\right)$ product rule

$$\begin{aligned} \frac{d}{dx} x^{\frac{1}{2}} \sec \frac{1}{x} \\ = \frac{1}{2} x^{-\frac{1}{2}} \sec \frac{1}{x} + x^{\frac{1}{2}} \sec \frac{1}{x} \tan \frac{1}{x} \left(-\frac{1}{x^2}\right) \end{aligned}$$

[3]

iv Use the quotient rule to find $f'(t)$ if $f(t) = \frac{1-4^t}{2+3t}$.

Gebruik die kwosiëntreël om $f'(t)$ te bepaal indien $f(t) = \frac{1-4^t}{2+3t}$

$$\begin{aligned} f'(t) &= \frac{d}{dt} \frac{1-4^t}{2+3t} \\ &= \frac{(-1)4^t \ln 4 (2+3t) - (1-4^t) \times 3}{(2+3t)^2} \end{aligned}$$

[3]

v Find / Bepaal $\frac{d^2}{dx^2} \cos 3x$

$$\begin{aligned} \frac{d^2}{dx^2} \cos 3x &= \frac{d}{dx} \left(\frac{d}{dx} \cos 3x \right) \\ &= \frac{d}{dx} (-3 \sin 3x) \\ &= -3 \times \cos 3x \times 3 \\ &= -9 \cos 3x \end{aligned}$$

[2]

Question 18 / Vraag 18

Consider the function / Beskou die funksie

$$f(x) = \begin{cases} a^2x - 2 & \text{if / as } x < 1 \\ \frac{a}{x} & \text{if / as } x > 1 \\ -1 & \text{if / as } x = 1 \end{cases}$$

- i Find the value(s) of a for which $\lim_{x \rightarrow 1} f(x)$ exists. Explain your answer.

Bepaal die waarde(s) van a waarvoor $\lim_{x \rightarrow 1} f(x)$ bestaan. Verduidelik u antwoord.

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{a}{x} = a$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (a^2x - 2) = a^2 - 2$$

$$\lim_{x \rightarrow 1} f(x) \text{ exists } \Leftrightarrow \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$$\Leftrightarrow a = a^2 - 2$$

$$\Leftrightarrow a^2 - a - 2 = (a-2)(a+1) = 0$$

$$\Leftrightarrow a = 2 \text{ or } a = -1$$

[3]

- ii Find the value(s) of a for which $y = f(x)$ will be continuous at $x = 1$. Explain your answer.

Bepaal die waarde(s) van a waarvoor $y = f(x)$ kontinu in $x = 1$ sal wees. Verduidelik u antwoord.

$$f \text{ is continuous at } x=1 \text{ if } f(1) = \lim_{x \rightarrow 1} f(x)$$

(1) $f(1) = -1$ is defined

$$\left. \begin{array}{l} \text{(2) if } a=2 \text{ then } \lim_{x \rightarrow 1} f(x) = 2 \\ \text{if } a=-1 \text{ then } \lim_{x \rightarrow 1} f(x) = -1 \end{array} \right\} \begin{array}{l} \lim_{x \rightarrow 1} f(x) \text{ exists} \\ \text{if } a=-1 \text{ or } a=2 \end{array}$$

[2]

(3) $f(1) = -1 = \lim_{x \rightarrow 1} f(x)$ only if $a = -1$

(NB) You cannot do (i) without referring to (i)

Question 19 / Vraag 19

- i Complete the definition: A function f is continuous at $x = a$ if ...

Voltooi die definisie: 'n Funksie f is kontinu in $x = a$ as ...

$$\lim_{x \rightarrow a} f(x) = f(a)$$

[1]

- ii According to the definition, a function f is differentiable at $x = a$ if a certain limit exists. You can write that limit in two ways. Write down both limits.

Volgens die definisie is 'n funksie f differensieerbaar in $x = a$ indien 'n spesifieke limiet bestaan. Die limiet kan op twee maniere neergeskryf word. Skryf beide limiete neer.

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists}$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists}$$

[2]

- iii Prove that if a function f is differentiable at $x = a$, then f is continuous at $x = a$.

Bewys dat indien 'n funksie differensieerbaar in $x = a$ is, dan is f kontinue in $x = a$.

$$\lim_{x \rightarrow a} [f(x) - f(a)]$$

$$= \lim_{x \rightarrow a} f(x) - f(a) \times \frac{x - a}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times \lim_{x \rightarrow a} (x - a)$$

$$= f'(a) \times 0 = 0$$

[2]

$$\therefore \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x) - f(a) + f(a)]$$

$$= \lim_{x \rightarrow a} [f(x) - f(a)] + f(a)$$

$$= f(a)$$